

Learning the Metric Induced by a G_2 -Structure

Generalisation and Equivariance

Project proposal — AI for Geometry (MM845)

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1. Mathematical problem

Geometric setting. A positive differential 3-form on a seven-dimensional manifold determines a Riemannian metric pointwise. This project studies that algebraic construction on an oriented vector space $V = \mathbb{R}^7$, rather than solving a differential equation or imposing torsion-free conditions [1]. Let $(e^1, \dots, e^7)$ be the standard dual basis, write $e^{ijk} = e^i \wedge e^j \wedge e^k$, and fix

$$\varphi_0 = e^{123} + e^{145} + e^{167} + e^{246} - e^{257} - e^{347} - e^{356}.$$

Its stabiliser is the compact exceptional group $G_2 \subset \mathrm{SO}(7)$. We work with the open orbit $\mathcal{P} = \{A^*\varphi_0 : A \in \mathrm{GL}^+(7, \mathbb{R})\}$ of positive forms compatible with the fixed orientation. Each $\varphi \in \mathcal{P}$ determines a unique positive-definite metric g_φ [2].

The target map. With $\mathrm{vol}_0 = e^1 \wedge \dots \wedge e^7$, define the symmetric matrix B_φ by

$$(B_\varphi)_{ij} \mathrm{vol}_0 = \frac{1}{6} (\iota_{e_i} \varphi) \wedge (\iota_{e_j} \varphi) \wedge \varphi, \quad g_\varphi = (\det B_\varphi)^{-1/9} B_\varphi.$$

Here ι_{e_i} denotes contraction with the basis vector e_i ; these signs agree with the convention in [2, Section 2]. The learning task is to approximate the nonlinear map

$$\mathcal{F} : \mathcal{P} \subset \Lambda^3 V^* \longrightarrow \mathrm{Sym}^+(7, \mathbb{R}), \quad \varphi \longmapsto g_\varphi.$$

In coordinates, the input has $\binom{7}{3} = 35$ coefficients and the output has $7 \cdot 8/2 = 28$ independent entries.

Geometric structure to test. Naturality gives, for every $C \in \mathrm{GL}^+(7, \mathbb{R})$, the change-of-basis identity

$$\mathcal{F}(C^*\varphi) = C^\top \mathcal{F}(\varphi) C.$$

In particular, $\mathcal{F}(A^*\varphi_0) = A^\top A$, since $g_{\varphi_0} = I$. Taking $C = t^{1/3}I$ also yields $\mathcal{F}(t\varphi) = t^{2/3}\mathcal{F}(\varphi)$ for $t > 0$. These identities provide independent tests of whether a learned map respects the geometry, beyond its accuracy on randomly held-out examples.

Research question and significance. Can a small neural network learn this map accurately, and does training with changes of basis improve its equivariance and generalisation to unseen scales and anisotropies? The formula is already known: the objective is neither to discover a new metric construction nor to claim a faster replacement for exact evaluation. Instead, it provides a controlled mathematical benchmark for comparing ordinary supervised regression with symmetry-aware data augmentation, motivated by [3]. The working hypothesis is that augmentation reduces equivariance error on unseen examples; prediction accuracy and extrapolation will be measured separately, without assuming that either improves.

2. AI setup and experimental plan

Data and generation. Generate an initial dataset of 30,000 pairs $(\varphi, g) = (A^* \varphi_0, A^\top A)$ in double precision, with $A = Q_1 \text{diag}(e^{s_1}, \dots, e^{s_7}) Q_2$, independent Haar-distributed $Q_1, Q_2 \in \text{SO}(7)$, and independent $s_i \sim \text{Unif}[-0.35, 0.35]$. This controls conditioning while varying both the form and metric; orthogonal matrices alone would give only $g = I$. Validate the generator against the exterior-algebra formula for B_φ and the case $A = I$. Labels are analytically known, with only floating-point error. The network receives the 35 coefficients φ_{ijk} ($i < j < k$), *not* the generating matrix A .

Splits and generalisation. Split the original pairs into 70% training, 15% validation and 15% test *before* augmentation; transformed copies inherit their source split. Fit input standardisation on training data only and do not discard the scale of φ . A separate extrapolation set will use $s_i \in [-0.7, 0.7]$ with $\max_i |s_i| > 0.35$, so that at least one singular value lies outside the training range. No test set will guide model selection.

Learning method and comparisons. Compare ridge regression on the 28 metric entries with a supervised multilayer perceptron (MLP), initially 35–128–128–28 with ReLU activations. The MLP outputs a lower-triangular matrix L , with softplus diagonal plus a small positive constant; its prediction is $\hat{g}_\theta = LL^\top$, ensuring positive definiteness. Train with Adam, validation-based early stopping, and mean squared relative Frobenius loss,

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{n=1}^N \frac{\|\hat{g}_\theta(\varphi_n) - g_n\|_F^2}{\|g_n\|_F^2}.$$

Train an otherwise identical MLP with online augmentation $(\varphi, g) \mapsto (C^* \varphi, C^\top g C)$ for random $C \in \text{SO}(7)$. Orthogonal augmentation preserves the singular-value range, keeping the extrapolation test separate. Match architectures, source data and maximum optimisation budgets. This tests learned, not architecturally guaranteed, equivariance; an exactly $\text{GL}^+(7, \mathbb{R})$ -equivariant architecture is outside scope.

Evaluation and success criteria. Report median and 95th-percentile relative metric error on both test sets, learning curves for increasing training-set sizes, and variation across three random seeds. Separately measure the equivariance defect

$$E_{\text{eq}}(\varphi, C) = \frac{\|\hat{g}_\theta(C^* \varphi) - C^\top \hat{g}_\theta(\varphi) C\|_F}{\|C^\top g_\varphi C\|_F}.$$

Test fresh rotations and bounded non-orthogonal transformations separately, and check the scaling law. Record positive-definiteness failures of the unconstrained linear baseline. Evidence for the hypothesis requires a reproducible reduction in equivariance error without a material loss of prediction accuracy; failure to extrapolate is also an informative outcome.

Feasibility and deliverables. The implementation will use Python/NumPy and PyTorch, with version control, fixed seeds and recorded configurations. Deliverables are reproducible code, validated synthetic data, error plots and a report explaining which geometric properties are learned and where the models fail. AI coding assistance will be documented, with mathematical identities and numerical tests checked independently.

References

- [1] S. Karigiannis. *Introduction to G_2 geometry*. In *Lectures and Surveys on G_2 -Manifolds and Related Topics*, Fields Institute Communications, vol. 84, Springer, 2020. arXiv:1909.09717.
- [2] S. Grigorian. *Deformations of G_2 -structures with torsion*. Asian Journal of Mathematics **20**(1), 123–155, 2016. arXiv:1108.2465.
- [3] S. Chen, E. Dobriban and J. H. Lee. *A Group-Theoretic Framework for Data Augmentation*. Journal of Machine Learning Research **21**(245), 1–71, 2020. Article 20-163.